

Subject: Programming with Java

ASSIGNMENT No : 4

TITLE :Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.

Problem Statement

Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.

Learning Objectives

To perform conversion between primitive data types and wrapper objects using Java wrapper classes.

Theory

Wrapper classes are predefined classes that encapsulate primitive data types into objects. Java provides wrapper classes such as Integer, Double, Character, and Boolean for every primitive type. They support autoboxing, unboxing, parsing, and conversion methods. Wrapper classes are widely used in Java Collections Framework and when object representation of primitive values is required.

Conclusion

Hence, the use of Java wrapper classes was successfully demonstrated to encapsulate primitive data types into objects and parse string inputs into numeric values. By implementing autoboxing, unboxing, and validation operations within practical scenarios like calculating student marks and processing employee payrolls, the vital role of wrapper classes in type conversion and object-oriented data management was effectively established.

Exercise

1. Create a program to convert student marks from String format to Integer and calculate total marks.

Code:

```
1  public class StudentMarks {  
    Run main | Debug main  
2      public static void main(String[] args) {  
3  
4          String mathStr = "88";  
5          String physicsStr = "92";  
6  
7          Integer mathMarks = Integer.valueOf(mathStr);  
8          Integer physicsMarks = Integer.valueOf(physicsStr);  
9  
10         int totalMarks = mathMarks + physicsMarks;  
11  
12         System.out.println("Name: Ali Mehdi Jafri, PRN: 25070122031");  
13         System.out.println("Math: " + mathMarks);  
14         System.out.println("Physics: " + physicsMarks);  
15         System.out.println("Total Marks: " + totalMarks);  
16     }  
17 }
```

Output:

```
Listening on 60709  
User program running  
User program finished  
Name: Ali Mehdi Jafri, PRN: 25070122031  
Math: 88  
Physics: 92  
Total Marks: 180
```

PTO

2. Develop an **Employee Payroll System** that accepts employee IDs, basic salary, and bonus amounts from the user. Convert the entered values into wrapper objects and perform validation operations to ensure valid salary values before calculating the net salary.

Code:

```
1 import java.util.Scanner;
2 public class EmployeePayroll {
3     Run main | Debug main
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Enter Employee ID: ");
8         String idInput = sc.nextLine();
9
10        System.out.print("Enter Basic Salary: ");
11        String salaryInput = sc.nextLine();
12
13        System.out.print("Enter Bonus: ");
14        String bonusInput = sc.nextLine();
15
16        try {
17            Integer empId = Integer.valueOf(idInput);
18            Double basicSalary = Double.valueOf(salaryInput);
19            Double bonus = Double.valueOf(bonusInput);
20            if (basicSalary > 0 && bonus >= 0) {
21                double netSalary = basicSalary + bonus;
22                System.out.println("\n--- Payroll Slip ---");
23                System.out.println("ID: " + empId);
24                System.out.println("Net Salary: $" + netSalary);
25            } else {
26                System.out.println("Error: Values must be positive.");
27            }
28        } catch (NumberFormatException e) {
29            System.out.println("Error: Invalid numeric format entered.");
30        }
31        sc.close();
32    }
33 }
```

Output:

```
Listening on 64421
User program running
Enter Employee ID:
→ 101

Enter Basic Salary:
→ 100000

Enter Bonus:
→ 10000

--- Payroll Slip ---
ID: 101
Net Salary: $110000.0
User program finished
```